

Post-Spotlight Posts: The Impact of Sudden Social Media Attention on Account Behavior

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ABSTRACT

Within social media communities, sudden attention is a frequent phenomenon where a particular user receives a rapid onset of heightened engagement. Using a statistical analysis, we examine the impacts that this outsized attention has on the subsequent behavior of accounts who received sudden social media attention for Twitter posts in the early stages of the Covid-19 pandemic. We find that accounts that received sudden social media attention were more likely to post after receiving sudden social media attention, though this effect was not persistent and did not boost original content production. We also find that these accounts were significantly more likely to alter their self-presentation through updating their “bio” after receiving suddenly increased attention.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing**; **Social networks**; **Social media**.

KEYWORDS

social media analysis; influencers; Twitter; Bayesian modeling

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1 INTRODUCTION AND BACKGROUND

Crisis events can cause a convergence of attention to events, as people attempt to make sense of confusing, high-stakes informational environments [8]. One common conversational dynamic in rapidly evolving information environments online is when accounts that have low levels of normal engagement suddenly receive disproportionate attention. This can occur for a variety of reasons, including ‘spotlighting’ of small accounts by influencers [17], or more general

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‘popularity shocks’ [6]. In this study, we examine Twitter posts during the early stages of the Covid-19 pandemic to see the effects of sudden social media attention.

Recent research has shown that social media and influencers (accounts, typically with large followings and/or dedicated audiences, who are significant forces in shaping online conversations) are critical information sources for many [3, 10]. Further, influencers can significantly impact what information spreads on social media [10, 12]. While not directly studying influencers, those who receive sudden social media attention are at least temporarily influential, and may be able to use this moment of attention to become influencers.

Other recent papers have begun to explore how audience feedback influences social media user behavior. Hasan et. al. researched how visibility impacted scientists’ behavior on Twitter [7], which found that scientists tended to tweet more topically similar content after going viral. Additionally, Adelani et. al. showed that social media users are somewhat influenced by community feedback on the topics they post about [1]. Gurjar et. al. found that on their platform of study, “popularity shocks” only changed user behavior in the short term, and that users had difficulty in getting increased attention to persist [6]. Both Gurjar et. al. and Hasan et. al. also found users tended to engage more with the platform after a popularity shock [6, 7]. Our study seeks to corroborate these studies’ findings about engagement, while also examining whether accounts make longer-lasting changes in their self-presentation (via their account description, also known as an account’s “bio”) after such an event.

2 QUESTIONS

Given the previous research on this topic, we identified three research questions:

RQ1 [Engagement]: Do Twitter accounts post more after receiving a sudden increase in attention?

RQ2 [Original Posts]: Do accounts post more original (non-retweet/reply) tweets after receiving sudden attention?

RQ3 [Self-Presentation]: After sudden attention, do accounts alter their self-presentation (their profile bio)?

3 METHODS

To study these questions, we used Bayesian modeling as described in van de Schoot et. al.’s methodological overview [15]. Specifically, we rely on a Bayesian model of a negative binomial distribution for **RQ1** and **RQ2**, since some accounts post extremely high amounts relative to the normal engagement level of accounts, and having

negative posts is impossible. We use a beta distribution for **RQ3**, since we are modeling a probability function.

To get data for the study, we used two APIs provided by Twitter. First, since January of 2020 the Center for an Informed Public has collected a database of Twitter posts relating to the coronavirus pandemic, using Twitter’s Streaming API. Using this data, we collected posts containing the case-insensitive substring “covid19” between April 15 and June 1, 2020. Next, we created a random control group sample, *random*, which consisted of 500 random tweets containing “covid19” in that time period. We then created a group of highly-retweeted tweets from accounts which typically get large amounts of attention, *popular*, consisting of the 500 most-retweeted tweets where the account’s lowest recorded follower account was greater than half of the number of retweets the post received. Finally, the population of most interest, *spotlight*, consisted of the 500 most-retweeted tweets where the number of retweets was over double the account’s minimum number of followers in the time interval. To make sure that the samples were roughly occurring in this time interval, at least 25% of the retweets of the post had to have occurred in the interval. We use two control groups here, to make sure behavioral shifts for the accounts in *spotlight* are different both compared to the general account population on Twitter and compared to other accounts that consistently get high engagement. Collectively, these tweets are what we describe as centered posts, as these correspond to either the posts which received sudden social media attention, or to the corresponding posts in our control groups.

While this data was sufficient for answering **RQ3**, we needed to collect more data to answer **RQ1** and **RQ2** since we did not have full platform engagement data for these accounts. In March and April 2022, we used Twitter’s Academic API to collect the complete public timelines of each account in these sample populations: 325 *popular* accounts, 490 *spotlight* accounts, and 499 *random* accounts (these numbers are less than 500 due to multiple tweets in a sample coming from the same account, and from accounts being deleted, suspended, or made private). Accounts were then filtered to remove accounts who posted at highly abnormal rates (over 200 tweets per day).

One limitation of this study is that we only observe accounts receiving sudden attention in Covid-19 discussions, so our findings may not apply to other contexts. Second, this study only focuses on Twitter, which limits applicability to other social media platforms, a challenge previously discussed by Zeynep Tufekci [14].

4 RESULTS

4.1 RQ1: Engagement

Using Bambi [4], we made a Bayesian, Negative Binomial generalized linear model to estimate the effect of receiving attention on posting frequency, controlling for activity prior to the centered tweet and variation in individual account tendencies, across the groups. We fit this model to our engagement data for time windows of 2 and 14 days before and after the centered posts, and calculate the measured behavior change of each group for **RQ1**. Note, in this and subsequent engagement models, we only include accounts which were active that number of days before and after.

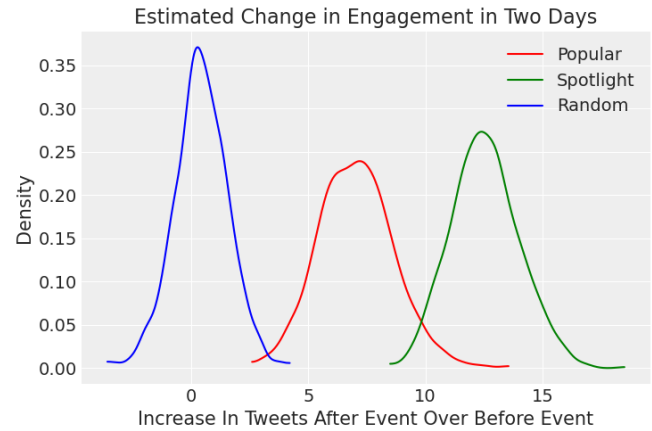


Figure 1: A density plot for the estimated increase in tweets in the two days after the centered tweet for accounts in each sample population compared to the two days prior to the centered post. This and subsequent density plots are generated with ArviZ [11]

Table 1: Measurements for modeled values of changes in net tweet frequency in the two days after the centered post for each sample compared to the two days prior.

Sample	Mean	Standard Deviation	CI-3%	CI-97%
<i>popular</i>	7.113	1.526	4.321	10.027
<i>spotlight</i>	12.56	1.513	9.903	15.533
<i>random</i>	0.368	1.11	-1.674	2.42

Table 2: Measurements for modeled values of changes in net tweet frequency in the fourteen days after the centered post for each sample compared to the fourteen days prior.

Sample	Mean	Standard Deviation	CI-3%	CI-97%
<i>popular</i>	18.81	6.767	6.858	31.835
<i>spotlight</i>	27.101	4.321	19.945	36.113
<i>random</i>	-1.006	5.23	-11.425	8.301

While (as shown in Figure 1 and Table 1) the 94% credible intervals for *popular* and *spotlight* overlap slightly, there is a large increase of both groups over *random* in platform engagement in the two days after the centering tweet. The mean for *spotlight* is also noticeably higher than *popular*. This shows receiving sudden increases in attention tends to increase short-term posting frequency.

Expanding the interval to 14 days before and after the centering post in Figure 2 and Table 2, we see a reduced effect. Since the time interval is multiplied by a factor of 7, we would expect the engagement difference to also be multiplied by 7 from the two day model if the effect persisted. Instead, we see an increase of roughly a factor of 2, likely indicating an approximate return to their prior baseline. This corroborates Gurjar et. al.’s study [6].

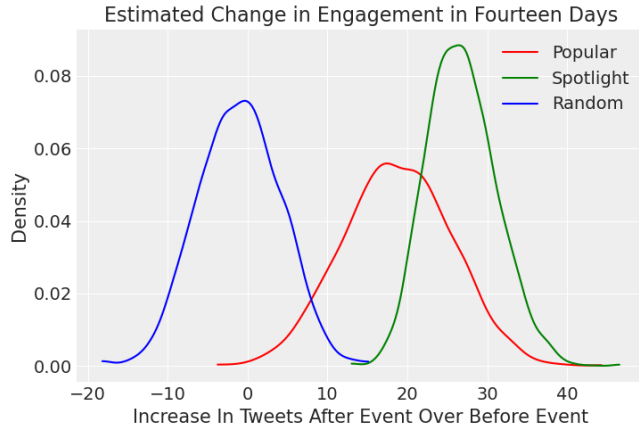


Figure 2: A density plot for the estimated increase in tweets in the fourteen days after the centered tweet for accounts in each sample population compared to the fourteen days prior to the centered tweet.

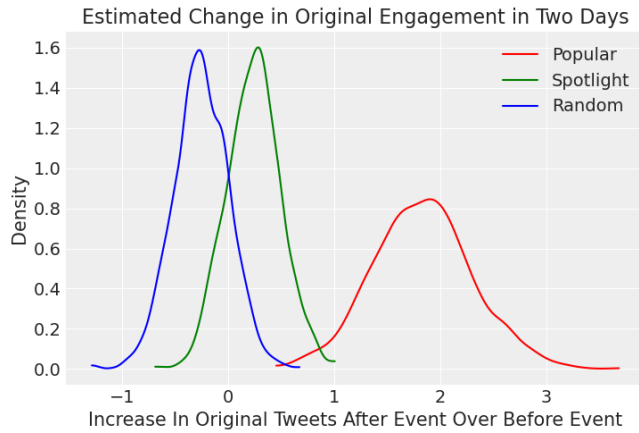


Figure 3: A density plot for the estimated increase in original tweets in the two days after the centered tweet for accounts in each sample population compared to the two days prior to the centered tweet.

4.2 RQ2: Original Engagement

To answer RQ2, we modeled the change in original posting rate (the amount of tweets made that are not retweets and not replies) in the two days before and after receiving a sudden increase in attention.

As shown in Figure 3 and Table 3, there is not a significant increase in original content posting frequency for accounts after a sudden attention increase. This indicates the majority of the earlier increase came from non-original engagement, i.e. replies or retweets.

To better understand what forms of non-original engagement increase, we also measure how much of this engagement is within the original conversation, by checking if the conversation ID of subsequent tweets matches the centered tweet. While over half

Table 3: Measurements for modeled values for the change in original tweeting frequency in the two days after the centered tweet compared to the two days prior.

Sample	Mean	Standard Deviation	CI-3%	CI-97%
<i>popular</i>	1.834	0.471	0.941	2.744
<i>spotlight</i>	0.238	0.254	-0.239	0.72
<i>random</i>	-0.256	0.26	-0.729	0.23

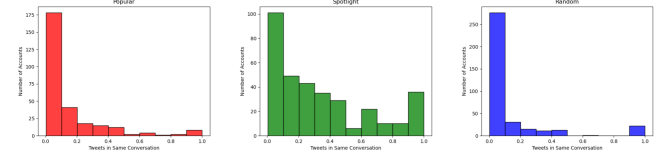


Figure 4: A histogram of users by subpopulation of the fraction of tweets in the centered conversation, made with matplotlib [9].

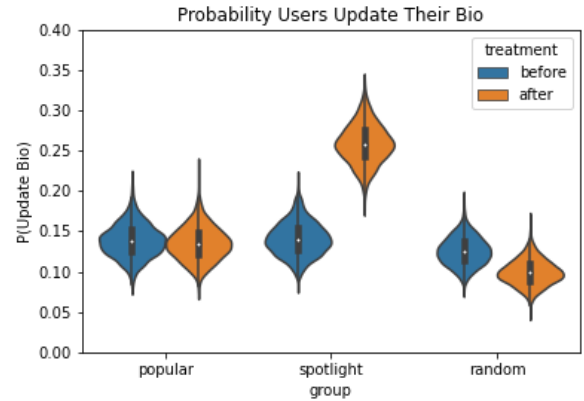


Figure 5: A violin plot of the estimated percentage of accounts in each group who on average would change their account self-description, or “bio,” in the two week periods before and after the centered tweet, made with Seaborn [16].

of tweets in the following two day period were in separate conversations for all groups on average, accounts in *spotlight* posted in the same conversation at a much higher rate (*popular*: 15.47% of tweets on average per-user; *random*: 13.34%; *spotlight*: 34.59%), further shown in Figure 4.

4.3 RQ3: Self-Description Changes

We then estimated the probability of a user updating their self-description before and after the centered post. Our model treated self-description changes in each group as binomially distributed, with a generic, weakly informative prior (Beta(2, 2)), and was trained with PyMC3 [13]. Note, we only include accounts with at least one description captured in the two-week period before and the two-week period after the event.

Table 4: Statistics for accounts changing their self-description in the two weeks before and two weeks after the centering post.

Sample	Total Number of Accounts	Changed Before	Changed After	Mean Change Difference Modeled
<i>popular</i>	277	37	36	-0.004
<i>spotlight</i>	300	41	77	0.118
<i>random</i>	368	45	35	-0.026
Sample	Standard Deviation	CI-3%	CI-97%	
<i>popular</i>	0.029	-0.062	0.045	
<i>spotlight</i>	0.031	0.059	0.177	
<i>random</i>	0.023	-0.072	0.014	

As shown in Figure 5 and Table 4, accounts in *spotlight* are significantly more likely to change their bio after the event than either control group, and this difference only occurs after the event. We augmented this approach by applying frequentist methods, yielding similar results. *Spotlight* is not significantly different from either control prior to the centered tweet (*popular-spotlight* chi-square statistic 0.0118, p-value .91; *spotlight-random* chi-square statistic .3049, p-value .58), but is significantly different after (*popular-spotlight* chi-square statistic 14.68, p-value .00013; *spotlight-random* chi-square statistic 30.91, p-value < .00001).

5 DISCUSSION

This study shows that accounts which see sudden large bursts of attention do have some behavioral shifts. They are far likelier to update their self-description in the two weeks after receiving this attention, and increase the amount they engage with the platform in the short term, primarily with non-original engagement and within the conversation receiving attention. More research is needed to understand what kinds of changes are made in self-presentation. Further, while this change in behavior might be transient for most accounts receiving sudden attention increases, focusing on those whose subsequent behavior is significantly altered should be a research priority.

Additionally, platform trace data as used here is useful for understanding how sudden attention correlates to account behavior changes, but it is insufficient for fully understanding impacts on users. People whose content has gained unexpectedly large amounts of attention on Twitter, originating both on and off the platform, have described the experience as disorienting and overwhelming [2, 5]. These anecdotal examples were written about as mostly unwelcome, or at least not positive. However, other accounts highlight when they are spotlighted by influential accounts, or when they more generally receive sudden attention. Understanding what makes some users celebrate or dislike this experience of their accounts receiving attention would be valuable for further research on these dynamics. More bluntly, this study's methods and findings are useful for studying *account* behaviors, as these can be measured

on the platform directly; it is less useful for understanding the impact of sudden attention on *users*, the actual people behind the accounts, as large-scale, quantitative methods can only measure their emotions and behaviors through proxy data and impressions that they leave through their digital traces.

Finally, looking longer-term at the impacts of sudden attention, particularly for accounts which maintain an increased level of influence, would be valuable for future research. When an account is engaged with by far more social media users, this necessarily exposes them to far larger audiences, which likely are meaningfully different from their prior audience. Understanding how users navigate these new communities could be extremely useful.

6 CONCLUSION

In this study, we examine the impact that sudden bursts of attention has on accounts' subsequent behavior on Twitter. We find that accounts are likely to post more on the platform after receiving increased attention, but only for a short amount of time and not for original content. Additionally, these accounts are more likely to update their self-presentation after receiving this attention. This helps researchers to better understand the impacts of these attention bursts on modern social media systems.

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REFERENCES

- [1] David Ifeoluwa Adelani, Ryota Kobayashi, Ingmar Weber, and Przemyslaw A. Grabowicz. 2020. Estimating community feedback effect on topic choice in social media with predictive modeling. *EPJ Data Science* 9, 1 (Dec. 2020), 1–23. <https://doi.org/10.1140/epjds/s13688-020-00243-w> Number: 1 Publisher: SpringerOpen.
- [2] Audrey Armstrong. 2022. Mamma Mia: On Accidentally Going Viral. <https://audreyarmstrongwriter.wordpress.com/2022/04/27/mamma-mia-on-accidentally-going-viral/>
- [3] Justin T. Brady, Molly E. Kelly, and Sharon L. Stein. 2017. The Trump Effect: With No Peer Review, How Do We Know What to Really Believe on Social Media? *Clinics in Colon and Rectal Surgery* 30, 4 (Sept. 2017), 270–276. <https://doi.org/10.1055/s-0037-1604256>
- [4] Tomás Capretto, Camen Pihó, Ravin Kumar, Jacob Westfall, Tal Yarkoni, and Osvaldo A. Martin. 2022. Bambi: A simple interface for fitting Bayesian linear models in Python. arXiv:2012.10754 [stat.CO]
- [5] Natalia Cecire. 2015. Everybody's Authority. *PMLA* 130, 2 (March 2015), 453–460. <https://doi.org/10.1632/pmla.2015.130.2.453> Publisher: Cambridge University Press.

- [6] Omkar Gurjar, Tanmay Bansal, Hitkul Jangra, Hemank Lamba, and Ponnuram Kumaraguru. 2022. Effect of Popularity Shocks on User Behaviour. *Proceedings of the International AAAI Conference on Web and Social Media* 16 (May 2022), 253–263. <https://doi.org/10.1609/icwsm.v16i1.19289>
- [7] Rakibul Hasan, Cristobal Cheyre, Yong-Yeol Ahn, Roberto Hoyle, and Apu Karpadia. 2022. The Impact of Viral Posts on Visibility and Behavior of Professionals: A Longitudinal Study of Scientists on Twitter. *Proceedings of the International AAAI Conference on Web and Social Media* 16 (May 2022), 323–334. <https://doi.org/10.1609/icwsm.v16i1.19295>
- [8] Amanda Lee Hughes and Leysia Palen. 2009. Twitter Adoption and Use in Crisis Twitter Adoption and Use in Mass Convergence and Emergency Events. In *Proceedings of the 6th International ISCRAM Conference*. Gothenburg, Sweden, 10 pages. <https://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.156.8385&rep=rep1&type=pdf>
- [9] John D. Hunter. 2007. Matplotlib: A 2D Graphics Environment. *Computing in Science & Engineering* 9, 3 (May 2007), 90–95. <https://doi.org/10.1109/MCSE.2007.55> Conference Name: Computing in Science & Engineering.
- [10] Ebuka Elias Igwebuike and Lily Chimunya. 2021. Legitimizing falsehood in social media: A discourse analysis of political fake news. *Discourse & Communication* 15, 1 (Feb. 2021), 42–58. <https://doi.org/10.1177/1750481320961659> Publisher: SAGE Publications.
- [11] Ravin Kumar, Colin Carroll, Ari Hartikainen, and Osvaldo Martin. 2019. ArviZ a unified library for exploratory analysis of Bayesian models in Python. *Journal of Open Source Software* 4, 33 (Jan. 2019), 1143. <https://doi.org/10.21105/joss.01143>
- [12] Zoe Lofft. 2020. When social media met nutrition: How influencers spread misinformation, and why we believe them. *Health Science Inquiry* 11, 1 (Aug. 2020), 56–61. <https://doi.org/10.29173/hsi319> Number: 1.
- [13] John Salvatier, Thomas V. Wiecki, and Christopher Fonnesbeck. 2016. Probabilistic programming in Python using PyMC3. *PeerJ Computer Science* 2 (April 2016), e55. <https://doi.org/10.7717/peerj-cs.55> Publisher: PeerJ Inc..
- [14] Zeynep Tufekci. 2014. Big Questions for Social Media Big Data: Representativeness, Validity and Other Methodological Pitfalls. *Proceedings of the International AAAI Conference on Web and Social Media* 8, 1 (May 2014), 505–514. <https://ojs.aaai.org/index.php/ICWSM/article/view/14517> Number: 1.
- [15] Rens van de Schoot, Sarah Depaoli, Ruth King, Bianca Kramer, Kaspar Märtens, Mahlet G. Tadesse, Marina Vannucci, Andrew Gelman, Duco Veen, Joukje Willemssen, and Christopher Yau. 2021. Bayesian statistics and modelling. *Nature Reviews Methods Primers* 1, 1 (Jan. 2021), 1–26. <https://doi.org/10.1038/s43586-020-00001-2> Number: 1 Publisher: Nature Publishing Group.
- [16] Michael L. Waskom. 2021. seaborn: statistical data visualization. *Journal of Open Source Software* 6, 60 (April 2021), 3021. <https://doi.org/10.21105/joss.03021>
- [17] Kaitlyn Zhou, Tom Wilson, Kate Starbird, and Emma S. Spiro. 2023. Spotlight Tweets: A Lens for Exploring Attention Dynamics within Online Sensemaking During Crisis Events. *Trans. Soc. Comput.* 6, 1–2, Article 2 (jun 2023), 33 pages. <https://doi.org/10.1145/3577213>

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